



上海冠显光电科技有限公司  
Shanghai Top Display Optoelectronics Co., LTD

## LCD MODULE SPECIFICATION

Customer: \_\_\_\_\_

Module No.: TL088PWV01-H1608B

Date: 2025-11-14

Version: 2.0

For Customer's Acceptance:

| Approved by | Comment |
|-------------|---------|
|             |         |

| Approved by   | Checked by   | Prepared by  |
|---------------|--------------|--------------|
| Kanglin.Zhong | Xianren.Zhou | Zaiping.Yang |

## Table of Contents

|                                            |    |
|--------------------------------------------|----|
| Record of Revision .....                   | 3  |
| 1 General Specifications .....             | 4  |
| 2 Pin Assignment .....                     | 5  |
| 3 Absolute Maximum Ratings .....           | 6  |
| 4. Electrical Characteristics .....        | 6  |
| 5 Timing Chart .....                       | 8  |
| 6 Optical Characteristics .....            | 13 |
| 7 Environmental / Reliability Test .....   | 16 |
| 8 Mechanical Drawing .....                 | 17 |
| 9 Precautions for Use of LCD Modules ..... | 18 |

## Record of Revision

| Rev. | Date       | Description                                                  | Editor        |
|------|------------|--------------------------------------------------------------|---------------|
| 1.0  | 2023-04-08 | First release                                                | Kanglin.Zhong |
| 2.0  | 2025-11-14 | Change the brightness settings in the specification document | Ruipan.Zhao   |
|      |            |                                                              |               |
|      |            |                                                              |               |
|      |            |                                                              |               |
|      |            |                                                              |               |
|      |            |                                                              |               |
|      |            |                                                              |               |
|      |            |                                                              |               |
|      |            |                                                              |               |
|      |            |                                                              |               |
|      |            |                                                              |               |
|      |            |                                                              |               |
|      |            |                                                              |               |
|      |            |                                                              |               |
|      |            |                                                              |               |
|      |            |                                                              |               |
|      |            |                                                              |               |
|      |            |                                                              |               |

## 1 General Specifications

| No. | Item                           | Specification                    | Remark |
|-----|--------------------------------|----------------------------------|--------|
| 1   | LCD Size                       | 8.88 inch (Diagonal)             |        |
| 2   | Driver Element                 | a-Si TFT active matrix           |        |
| 3   | Resolution                     | 480 (RGB) ×1920                  |        |
| 4   | Display Mode                   | Normally Black, Transmissive     |        |
| 5   | Pixel Pitch(mm)                | 0.038 (H) × 0.114 (V)            |        |
| 6   | Display Colors                 | 16.7M                            |        |
| 7   | Surface Treatment              | --                               |        |
| 8   | Color Arrangement              | RGB-Stripe                       |        |
| 9   | Interface                      | MIPI                             |        |
| 10  | Viewing Direction              | All                              |        |
| 11  | Gray Scale Inversion Direction | /                                | Note 1 |
| 12  | Outline Dimension (mm)         | 64.05 (W) × 231.1 (H) × 4.75 (T) |        |
| 13  | Active Area (mm)               | 54.72(W) × 218.88 (H)            |        |
| 14  | Touch Screen                   | Without CTP                      |        |
| 15  | Display Driver IC              | OTA7290N                         |        |
| 16  | Touch Driver IC                | --                               |        |

Note 1: Viewing direction for best image quality is different from TFT definition. There is a 180°shift.

Note 2: RoHS compliant.

## 2 Pin Assignment

### 2.1 LCD Pin assignment

Match connector : 0.5mm pitch FPC/FFC,T=0.3mm.

Type of connector on FPC: XF2M-4015-1A (OMRON) or equivalent.

| N  | Symbol    | I/O | Description                                   | Remark |
|----|-----------|-----|-----------------------------------------------|--------|
| 1  | GND       | P   | Ground                                        |        |
| 2  | D0P       | I/O | MIPI DSI differential data pair (Data lane 0) |        |
| 3  | D0N       | I/O | MIPI DSI differential data pair (Data lane 0) |        |
| 4  | GND       | P   | Ground                                        |        |
| 5  | D1P       | I   | MIPI DSI differential data pair (Data lane 1) |        |
| 6  | D1N       | I   | MIPI DSI differential data pair (Data lane 1) |        |
| 7  | GND       | P   | Ground                                        |        |
| 8  | DCKP      | I   | MIPI DSI differential clock pair              |        |
| 9  | DCKN      | I   | MIPI DSI differential clock pair              |        |
| 10 | GND       | P   | Ground                                        |        |
| 11 | D2P       | I   | MIPI DSI differential data pair (Data lane 2) |        |
| 12 | D2N       | I   | MIPI DSI differential data pair (Data lane 2) |        |
| 13 | GND       | P   | Ground                                        |        |
| 14 | D3P       | I   | MIPI DSI differential data pair (Data lane 3) |        |
| 15 | D3N       | I   | MIPI DSI differential data pair (Data lane 3) |        |
| 16 | GND       | P   | Ground                                        |        |
| 17 | GND       | P   | Ground                                        |        |
| 18 | VCC_1V8   | --  | Not connect                                   |        |
| 19 | VCC_1V8   | --  | Not connect                                   |        |
| 20 | REFCLK/NC | --  | Not connect                                   |        |
| 21 | SDA       | --  | Not connect                                   |        |
| 22 | SCL       | --  | Not connect                                   |        |
| 23 | BIST      | --  | Not connect                                   |        |
| 24 | RSTB      | I   | Reset Signal ,Active Low(1.8V)                |        |
| 25 | STBYB     | --  | Not connect                                   |        |
| 26 | AVDD/NC   | --  | Not connect                                   |        |
| 27 | GND       | P   | Ground                                        |        |
| 28 | K         | P   | LED CATHODE                                   |        |
| 29 | K         | P   | LED CATHODE                                   |        |
| 30 | GND       | P   | Ground                                        |        |
| 31 | VGL/NC    | --  | Not connect                                   |        |
| 32 | GND       | P   | Ground                                        |        |

|    |         |    |                                |  |
|----|---------|----|--------------------------------|--|
| 33 | GND     | P  | Ground                         |  |
| 34 | VGH/NC  | -- | Not connect                    |  |
| 35 | A       | P  | LED ANODE                      |  |
| 36 | A       | P  | LED ANODE                      |  |
| 37 | GND     | P  | Ground                         |  |
| 38 | VDD-3V3 | P  | LCD analog power supply (3.3V) |  |
| 39 | VDD-3V3 | P  | LCD analog power supply (3.3V) |  |
| 40 | VCOM/NC | -- | Not connect                    |  |

I---Input, O---Output, P--- Power/Ground

### 3 Absolute Maximum Ratings

Ta = 25℃

| Item                           | Symbol           | Min.  | Max. | Unit  | Remark |
|--------------------------------|------------------|-------|------|-------|--------|
| Power Voltage                  | VDD-3V3          | -0.30 | +4.5 | V     |        |
|                                | VCC_1V8          | -     | -    | V     |        |
| Operating Temperature          | Top              | -20.0 | 70.0 | ℃     |        |
| Storage Temperature            | T <sub>st</sub>  | -30.0 | 80.0 | ℃     |        |
| Operating and Storage Humidity | H <sub>stg</sub> | 10%   | 90%  | %(RH) |        |

### 4. Electrical Characteristics

#### 4.1 Recommended Operating Condition

VDD-3V3, GND=0V, Ta = 25℃

| Item                              | Symbol                        | Min.             | Typ. | Max.             | Unit | Remark                          |
|-----------------------------------|-------------------------------|------------------|------|------------------|------|---------------------------------|
| Digital supply Voltage            | VCC_1V8                       | --               | --   | --               | V    |                                 |
| Analog supply Voltage             | VDD-3V3                       | 2.8              | 3.0  | 3.3              | V    |                                 |
| TP Power                          | -                             | -                | -    | -                | V    |                                 |
| Input Signal Voltage              | Low Level<br>V <sub>IL</sub>  | 0                | -    | 0.3 x<br>VDD_3V3 | V    |                                 |
|                                   | High Level<br>V <sub>IH</sub> | 0.7 x<br>VDD_3V3 | -    | VDD_3V3          | V    |                                 |
| Current of digital supply voltage | I <sub>VCC_1V8</sub>          | -                | -    | -                | mA   |                                 |
| Current of analog supply voltage  | I <sub>VDD-3V3</sub>          | -                | 150  | 180              | mA   | VDD-3V3=3.3V, color bar pattern |

## 4.2 Backlight Unit Driving Condition

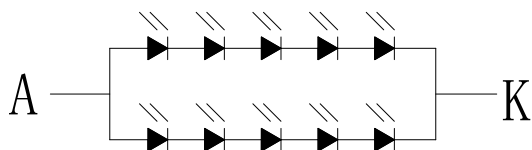
| Item                        | Symbol   | Min.  | Typ. | Max. | Unit | Remark                                    |
|-----------------------------|----------|-------|------|------|------|-------------------------------------------|
| Forward Current             | $I_F$    | -     | 180  | 200  | mA   | 10 LEDs<br>(5 LED Serial, 2 LED Parallel) |
| Forward Current Voltage     | $V_F$    | -     | 15   | 17.5 | V    |                                           |
| Backlight Power Consumption | $W_{BL}$ | -     | 2700 | 3500 | mW   |                                           |
| Operating Life Time         | --       | 20000 | --   | --   | hrs  | Note 2, Note 3                            |

Note1: The LED driving condition is defined for each module (5 LED Serial, 2 LED Parallel).

Note2: When LCM is operated, the stable forward current should be inputted. And forward voltage is for reference only.

Note3: Optical performance should be evaluated at  $T_a=25^{\circ}\text{C}$  When LED is driven at high current, high ambient temperature & humidity condition. The life time of LED will be reduced. Operating life means brightness goes down to 50% initial brightness. Typical operating life time is estimated data.

Note4: The LED driving condition is defined for each LED module.



## 5 Timing Chart

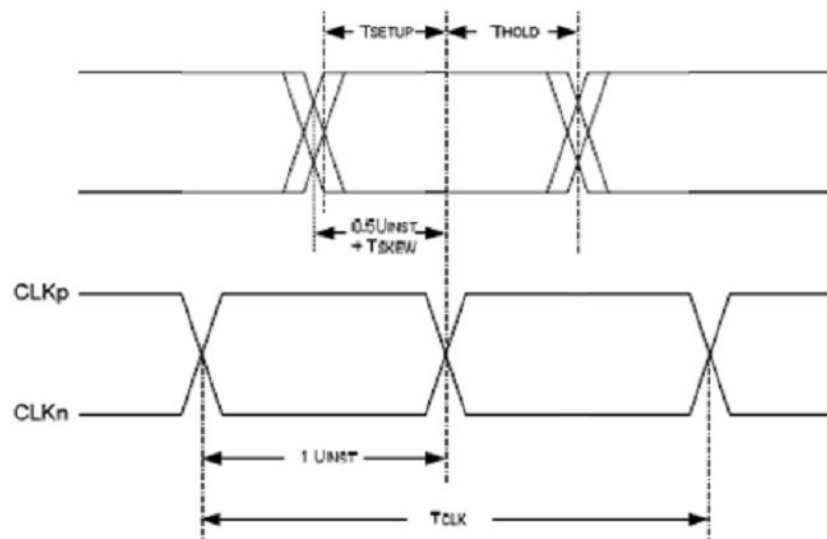
### 5.1 DSI Interface Timing Characteristics

#### HS Receiver AC Timing Characteristics

| Parameter                              | Symbol                | Rating |     |                | Unit        | Note                                                                                            |
|----------------------------------------|-----------------------|--------|-----|----------------|-------------|-------------------------------------------------------------------------------------------------|
|                                        |                       | Min    | Typ | Max            |             |                                                                                                 |
| Bandwidth per lane                     | -                     | -      | -   | 1000           | Mbps        | Bandwidth selected by register 'speedup'<br>Speedup=0 → Max=550Mbps<br>Speedup=1 → Max=1000Mbps |
| Operation frequency                    | -                     | -      | -   | 500            | MHz         |                                                                                                 |
| UI instantaneous                       | $UI_{INST}$           | 1      | -   | 12.5           | ns          | 1                                                                                               |
| Data to Clock Skew                     | $T_{skew}$            | -0.15  | -   | 0.15           | $UI_{INST}$ |                                                                                                 |
| Inter-lane static skew                 | $T_{skew-lane}$       | -      | -   | $UI_{INST}/50$ | $UI_{INST}$ |                                                                                                 |
| Data to Clock Setup Time               | $T_{SETUP}$           | 0.25   | -   | -              | $UI_{INST}$ | 2                                                                                               |
| Data to Clock Hold Time                | $T_{HOLD}$            | 0.25   | -   | -              | $UI_{INST}$ |                                                                                                 |
| Common-mode interference beyond 450MHz | $\Delta V_{CMRX(HF)}$ | -      | -   | 100            | mV          | 4                                                                                               |
| Common-mode interference 50MHz- 450MHz | $\Delta V_{CMRX(LF)}$ | -50    | -   | 50             | mV          | 3,6                                                                                             |
| Common-mode termination                | $C_{CM}$              | -      | -   | 60             | pF          | 5                                                                                               |

Note:

- (1) Total silicon and package delay budget of  $0.3 \cdot UI_{INST}$
- (2) Total setup and hold window for receiver of  $0.3 \cdot UI_{INST}$
- (3) Excluding 'static' ground shift of 50mV
- (4)  $\Delta V_{CMRX} (HF)$  is the peak amplitude of a sine wave superimposed on the receiver input
- (5) For higher bit rates a 14pF capacitor will be needed to meet the common-mode return loss specification.
- (6) Voltage difference compared to the DC average common-mode potential.



## 5.2 Recommended Timing Setting of TCON

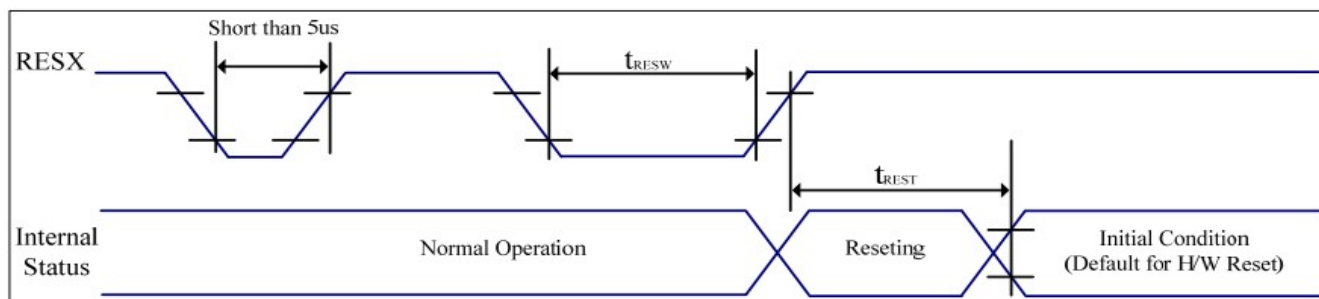
TCON (Embedded in Source IC) Input Timing (DCLK, HS, VS, DE)

VDD-3V3=3.3V, GND=0V, Ta=25°C

| Parameter | Symbol | Min. | Typ. | Max. | Unit | Remark |
|-----------|--------|------|------|------|------|--------|
| DCLK      | Fclk   | -    | 62   | -    | MHz  |        |
|           | tclk   | -    | 16.1 | -    | ns   |        |
| HSD       | thd    | -    | 480  | -    | tclk |        |
|           | thpw   | -    | 6    | -    | tclk |        |
|           | thb    | -    | 20   | -    | tclk |        |
|           | thfp   | -    | 30   | -    | tclk |        |
| VSD       | tvd    | -    | 1920 | -    | th   |        |
|           | tvpw   | -    | 6    | -    | th   |        |
|           | tvb    | -    | 10   | -    | th   |        |
|           | tvfp   | -    | 20   | -    | th   |        |

Note: For reference only, it needs to be adjusted according to the actual display effect.

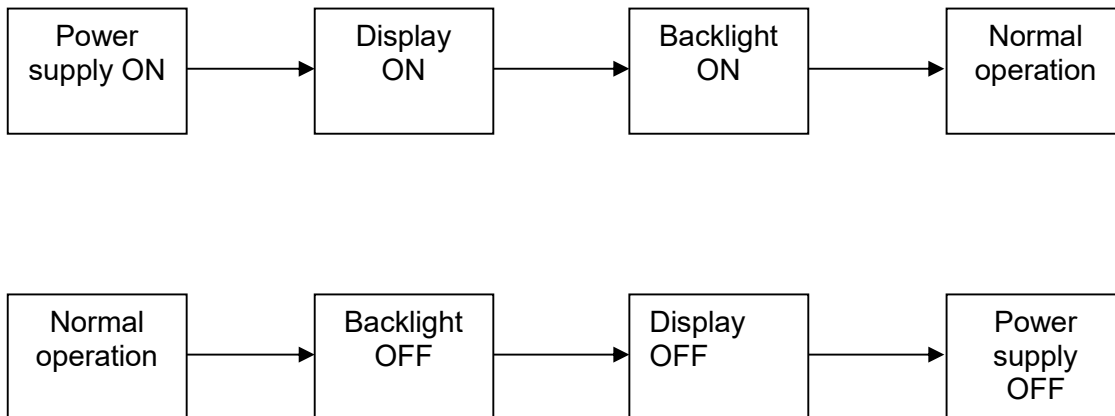
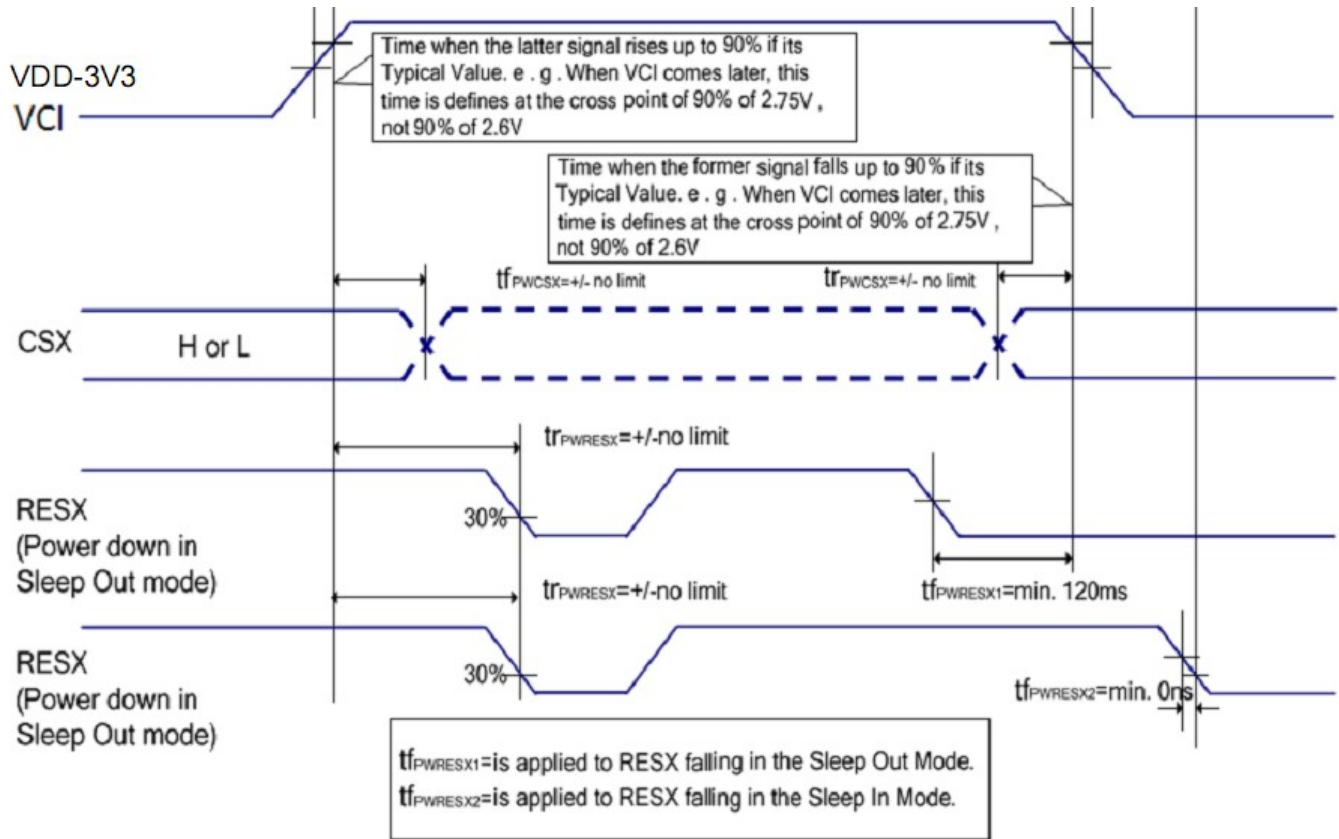
## 5.3 Reset input timing



| Symbol     | Parameter                 | Related Pins | MIN | TYP | MAX | Note                                     | Unit |
|------------|---------------------------|--------------|-----|-----|-----|------------------------------------------|------|
| $t_{RESW}$ | *1) Reset low pulse width | RESX         | 10  | -   | -   | -                                        | us   |
| $t_{REST}$ | *2) Reset complete time   | -            | -   | -   | 5   | When reset applied during Sleep in mode  | ms   |
|            |                           | -            | -   | -   | 120 | When reset applied during Sleep out mode | ms   |

Table: Reset input timing

## 5.4 Power On/Off Timing



## 6 Optical Characteristics

Ta=25℃

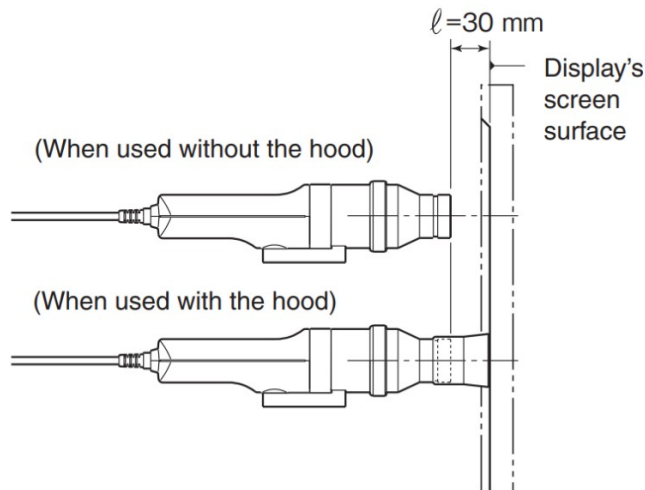
| Item           | Symbol     | Condition        | Min.            | Typ.  | Max.  | Unit              | Remark         |
|----------------|------------|------------------|-----------------|-------|-------|-------------------|----------------|
| View Angles    | $\theta T$ | $CR \geq 10$     | 70              | 80    | -     | Degree            | Note 2         |
|                | $\theta B$ |                  | 70              | 80    | -     |                   |                |
|                | $\theta L$ |                  | 70              | 80    | -     |                   |                |
|                | $\theta R$ |                  | 70              | 80    | -     |                   |                |
| Contrast Ratio | CR         | $\theta=0^\circ$ | 640             | 800   | -     |                   | Note1<br>Note3 |
| Response Time  | $T_{ON}$   | 25℃              | --              | 40    | -     | ms                | Note1<br>Note4 |
|                | $T_{OFF}$  |                  |                 |       |       |                   |                |
| Chromaticity   | White      | x                | Backlight is on | 0.246 | 0.296 | 0.346             | Note1<br>Note5 |
|                |            | y                |                 | 0.274 | 0.324 | 0.374             |                |
|                | Red        | x                |                 | 0.574 | 0.624 | 0.674             |                |
|                |            | y                |                 | 0.279 | 0.329 | 0.379             |                |
|                | Green      | x                |                 | 0.238 | 0.288 | 0.338             |                |
|                |            | y                |                 | 0.472 | 0.522 | 0.572             |                |
|                | Blue       | x                |                 | 0.086 | 0.136 | 0.186             |                |
|                |            | y                |                 | 0.087 | 0.137 | 0.187             |                |
| Uniformity     | U          |                  | 75              | 80    | -     | %                 | Note1<br>Note6 |
| NTSC           |            |                  | -               | 50    | -     | %                 | Note 5         |
| Luminance      | L          |                  | 540             | 600   | -     | cd/m <sup>2</sup> | Note1<br>Note7 |

Test Conditions:

1.  $I_F = 180 \text{ mA}$ ,  $V_F = 15 \text{ V}$  and the ambient temperature is  $25 \pm 2^\circ\text{C}$ . humidity is  $65 \pm 7\%$
2. The test systems refer to Note 1 and Note 2.

### Note 1: Definition of optical measurement system.

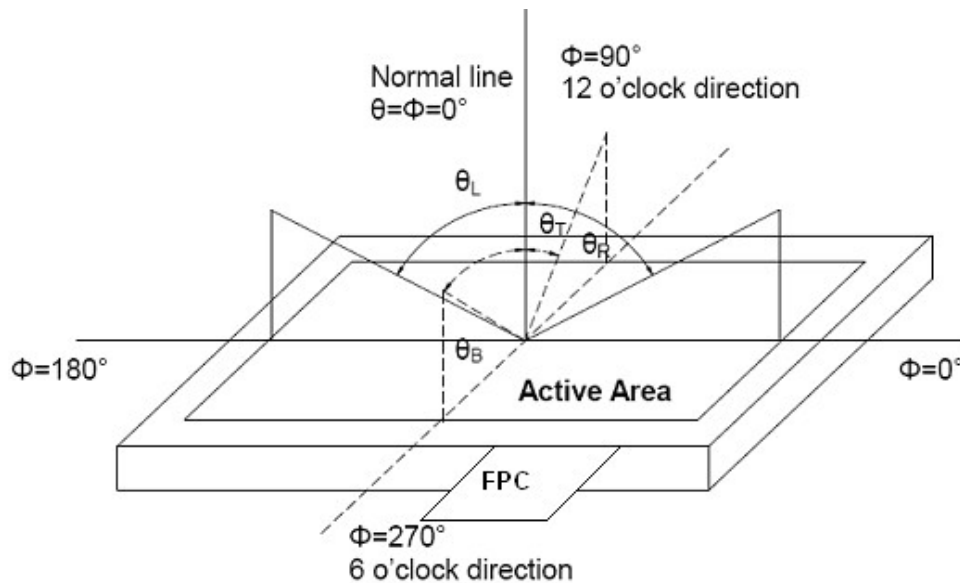
Properties are measured at the center point of the LCD screen. All input terminals LCD panel must be ground when measuring the center area of the panel.



| Item           | Photo detector             | Field |
|----------------|----------------------------|-------|
| Contrast Ratio | CA210 or similar equipment | 1°    |
| Luminance      |                            |       |
| Chromaticity   |                            |       |
| Lum Uniformity |                            |       |
| Response Time  | BM-7A                      | 2°    |

### Note 2: Definition of viewing angle range and measurement system.

Viewing angle is measured at the center point of the LCD by CONOSCOPE(ergo-80).



### Note 3: Definition of contrast ratio

$$\text{Contrast ratio (CR)} = \frac{\text{Luminance measured when LCD is on the "White" state}}{\text{Luminance measured when LCD is on the "Black" state}}$$

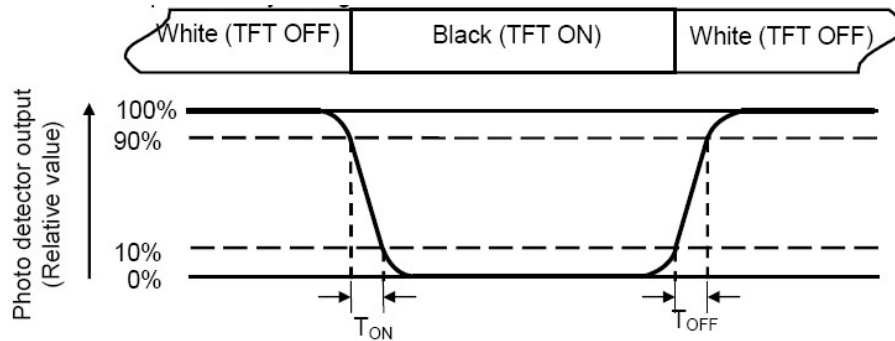
"White state ": The state is that the LCD should drive by V<sub>white</sub>.

"Black state": The state is that the LCD should drive by V<sub>black</sub>.

V<sub>white</sub>: To be determined      V<sub>black</sub>: To be determined.

#### Note 4: Definition of response time

The response time is defined as the LCD optical switching time interval between “White” state and “Black” state. Rise time ( $T_{ON}$ ) is the time between photo detector output intensity changed from 90% to 10%. And fall time ( $T_{OFF}$ ) is the time between photo detector output intensity changed from 10% to 90%.



#### Note 5: Definition of color chromaticity (CIE1931)

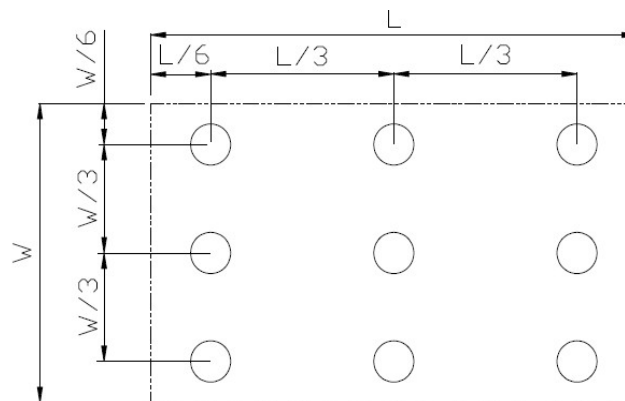
Color coordinates measured at center point of LCD.

#### Note 6: Definition of luminance uniformity

Active area is divided into 9 measuring areas (Refer Fig. 2). Every measuring point is placed at the center of each measuring area.

$$\text{Luminance Uniformity (U)} = L_{\min} / L_{\max}$$

L-----Active area length W----- Active area width



$L_{\max}$ : The measured Maximum luminance of all measurement position.

$L_{\min}$ : The measured Minimum luminance of all measurement position.

#### Note 7: Definition of luminance:

Measure the luminance of white state at center point.

## 7 Environmental / Reliability Test

| No | Test Item                                | Condition                                                                                                                              | Remarks                                                    |
|----|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| 1  | High Temperature Operation               | Ts = +70℃, 240 hours                                                                                                                   | No abnormalities in functions                              |
| 2  | Low Temperature Operation                | Ta = -20℃, 240 hours                                                                                                                   | No abnormalities in functions                              |
| 3  | High Temperature Storage                 | Ta = +80℃, 240 hours                                                                                                                   | No abnormalities in functions                              |
| 4  | Low Temperature Storage                  | Ta = -30℃, 240 hours                                                                                                                   | No abnormalities in functions                              |
| 5  | Storage at High Temperature and Humidity | Ta = +60℃, 90% RH max, 240 hours                                                                                                       | No abnormalities in functions                              |
| 6  | Thermal Shock (non-operating)            | -30℃ 30 min ~ +70℃ 30 min,<br>Change time: 0.5 hour ← 5 min → 0.5 hour. 10 Cycle                                                       | Start with cold temperature,<br>End with high temperature, |
| 7  | ESD                                      | C=150pF, R=330Ω, 5 point/panel<br>Air: ±8Kv, 5 times;<br>Contact: ±4Kv, 5 times<br>(Environment: 15℃~35℃,<br>30%~60% RH, 86Kpa~106Kpa) | No abnormalities in functions                              |

Note1: Ts is the temperature of panel's surface.

Note2: Ta is the ambient temperature of samples.

## 8 Mechanical Drawing



## 9 Precautions for Use of LCD Modules

### Handling Precautions

9.1.1 The display panel is made of glass. Do not subject it to a mechanical shock by dropping it from a high place, etc.

9.1.2 If the display panel is damaged and the liquid crystal substance inside it leaks out, be sure not to get any in your mouth, if the substance comes into contact with your skin or clothes, promptly wash it off using soap and water.

9.1.3 Do not apply excessive force to the display surface or the adjoining areas since this may cause the color tone to vary.

9.1.4 The polarizer covering the display surface of the LCD module is soft and easily scratched. Handle this polarizer carefully.

9.1.5 If the display surface is contaminated, breathe on the surface and gently wipe it with a soft dry cloth. If still not completely clear, moisten cloth with one of the following solvents:

- Isopropyl alcohol
- Ethyl alcohol

Solvents other than those mentioned above may damage the polarizer. Especially, do not use the following:

- Water
- Ketone
- Aromatic solvents

9.1.6 Do not attempt to disassemble the LCD Module.

9.1.7 If the logic circuit power is off, do not apply the input signals.

9.1.8 To prevent destruction of the elements by static electricity, be careful to maintain an optimum work environment.

9.1.8.1 Be sure to ground the body when handling the LCD Modules.

9.1.8.2 Tools required for assembly, such as soldering irons, must be properly ground.

9.1.8.3 To reduce the amount of static electricity generated, do not conduct assembly and other work under dry conditions.

9.1.8.4 The LCD Module is coated with a film to protect the display surface. Be care when peeling off this protective film since static electricity may be generated.

### Storage Precautions

9.2.1 When storing the LCD modules, avoid exposure to direct sunlight or to the light of fluorescent lamps.

9.2.2 The LCD modules should be stored under the storage temperature range. If the LCD modules will be stored for a long time, the recommend condition is: Temperature : 0°C ~ 40°C      Relatively humidity: ≤80%

9.2.3 The LCD modules should be stored in the room without acid, alkali and harmful gas.

### Transportation Precautions

9.3.1 The LCD modules should be no falling and violent shocking during transportation, and also should avoid excessive press, water, damp and sunshine.